MILITARY SPECIFICATION
BARRELS, TEST, SMALL ARMS AMMUNITION
GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers general requirements for test barrels used for proof testing of small arms ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

STANDARDS

Federal

Federal Test Method Standard No. 151	-	Metals: Test Method

Military

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material - Material shall be in accordance with the applicable barrel drawing.

3.2 Surface Finish - The surfaces shall be finished to the requirements indicated by symbols on the applicable barrel drawing.

3.3 Hardness - Hardness shall be in accordance with the applicable barrel drawing.

3.4 Physical Properties - Physical properties required by the drawing shall be uniform throughout each barrel within specified tolerances. Physical properties shall be achieved by quenching and tempering.

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3.5 Protective Finish - Protective finish shall be in accordance with the applicable barrel drawing and referenced specification and shall apply to exterior only.

3.6 Dimensional Requirements - Barrels shall comply with the dimensional requirements specified on the applicable drawing, (see 6.2 and 6.3).

3.7 Rifling Twist Deviation - Unless otherwise specified, lead of rifling shall not vary beyond 2 1/2 percent (2½%).

3.8 Bore Condition - Edges of lands shall be sharp and free from burrs.

3.9 Marking Requirements - Marking shall include the manufacturer's name or identification symbol, year of manufacture, part number and serial number. Marking shall be produced by impression stamping, engraving, etching, or other approved means. The marking shall be permanent, legible and logically grouped.

3.10 Workmanship - The workmanship shall be of the highest quality consistent with the tolerances and finishes specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions - Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection - Inspection shall be in accordance with referenced documents and as specified herein.

4.2 Inspection Provisions

4.2.1 Submission of Product - Barrels shall be submitted for acceptance, either at destination or the contractor's facility as required by the contract.

4.2.2 Examination

4.2.2.1 Dimensions - Each barrel shall be measured for all dimensions specified on the applicable barrel drawing. All non-conforming material shall be rejected.

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4.2.2.1.1 Rifling - The diameters of the lands and grooves shall be measured using an approved air gage. These measurements shall be made at one inch (1") intervals for each pair of lands and each pair of grooves. These diameters shall be recorded on the statement of findings for each barrel.

4.2.2.1.2 Chamber Contour - Chamber cast of "cerrolow 117" or equal may be made of barrels for determining dimensional requirements.

4.2.2.2 Workmanship and Condition - Each barrel shall be examined visually for satisfactory appearance of protective finish and for seams, cracks, nicks, scratches, rust spots and other defects which may materially affect the useability of the barrel. Rifling shall be visually inspected using an approved borescope. Rifling shall not have any scratches, tool marks or any irregularities. All non-conforming material shall be rejected.

4.2.3 Tests - The following tests shall be performed in accordance with 4.4.

4.2.3.1 Surface Finish - Each barrel shall be tested for surface finish. All non-conforming material shall be rejected.

4.2.3.2 Hardness - Barrels shall be tested for hardness. Failure of any barrel to comply with the hardness requirements of the applicable drawing shall be cause for rejection.

4.2.3.4 Tension Test - A minimum of two tensile specimens from each lot of barrels shall be tested for mechanical properties indicated on the applicable barrel drawing and certification of results shall be furnished. Failure of anyspecimen to comply with drawing requirements shall be cause for rejection of the lot of barrels.

4.2.4 Statement of Findings - A statement of findings shall be submitted for each lot of barrels presented to the Government for acceptance. Each statement shall include the following data:

a. Purchase order or contract number.
b. Item nomenclature and part number.
c. Applicable drawings and specifications used for inspection.
d. Method of inspection.
e. Authorized deviations, if any.
f. A statement of findings shall cover each barrel presented for acceptance.
g. Statement that inspection has been conducted in accordance with above criteria and that items have been found to conform with all requirements.
h. Signature of a responsible officer of the contractor.

4.2.5 Government Inspection and Acceptance - Barrels will be accepted

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based on an acceptable statement of findings. The Government, however, may conduct independent inspection to verify contractor findings. Such inspections shall be made at destination and non-conforming barrels shall be rejected and returned to the contractor. Such barrels may be reworked and resubmitted for acceptance.

4.2.6 Measuring and Test Equipment - The contractor shall provide and maintain measuring and test devices necessary to assure that barrels conform to technical requirements. In order to assure continued accuracy, these devices shall be calibrated at established intervals against certified standards which have known valid relationships to national standards. When required, the contractor's measuring and test equipment shall be made available for use by the Government to determine conformance of product with contract requirements. In addition, if conditions warrant, contractor's personnel shall be available for operation of such devices and for verification of their accuracy and condition.

4.3 Test Methods and Procedures

4.3.1 Surface Finish Test - The actual surface roughness shall be determined by measurement or by comparison of the surface to an acceptable standard reference surface.

4.3.2 Hardness - A hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

4.3.3 Tension - The criteria for performing this test shall be as prescribed by Federal Test Method Standard No. 151.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Barrels shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Marking - Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Ordering Data - Procurement documents should specify the title, number and date of this specification.

6.2 Reference Dimensions - A dimension labeled "Ref" is a non-mandatory dimension without tolerance, entered on the drawing for informational purposes only and does not directly govern machining or acceptance inspection.

6.3 Should additional dimensions and tolerances be required, application therefore shall be made through the Contracting Officer. Where

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dimensional tolerances are not indicated on the drawings by either a general tolerance note, or by specific tolerance limits, and the contractor had not applied for clarification prior to contract award, tolerances will be determined by the Government based upon essential functional and assembly requirements.

Custodian:
Army-MU

Preparing Activity:
Army-MU
Project No. 1305-A617

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Department of the Army

Frankford Arsenal
Philadelphia, Pa. 19137

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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